

PAPER_1679_HORIZON_60_EFOLDS

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PAPER_1679 — Inflation e-Fold Count = $N_{\text{efolds}} = A_5 = 60$ (minimum for horizon)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Inflation

Date: June 18, 2026

Status: CLOSED — EXACT closure (PAPER_145x-147x broader corpus)

Observation

PAPER_1462 (PAPER_14xx broader corpus, classic cosmology/BSM puzzle) gives:

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$N_{\text{efolds}} = A_5 = 60$ (minimum for horizon)

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Status: **EXACT**.

UQFF Closed Identity

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$N_{\text{efolds}} = A_5 = 60$ (minimum for horizon) EXACT

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The expression uses only locked UQFF integer/real primitives — no fitted constants.

Inflation e-Folds — Why 60?

Standard inflation requires ~60 e-folds of exponential expansion to solve the horizon problem (homogeneity of CMB across causally-disconnected regions). The number 60 has historically been empirical, fixed by matching to observed flatness and isotropy bounds.

UQFF supplies the structural reason:

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$N_{\text{efolds}} = A_5 = 60$ EXACT

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The icosahedral group order A_5 is also:

- Hayflick cell-division limit
- Heart rate (with $+SO_5$)
- SH0ES Hubble (with $+SO_5$)
- T_SCm activation temperature (K)
- Pop III IMF upper bound ($\times 2$)
- Quantum supremacy threshold (qubits)
- BH seed mass ($\times D_{\text{BSFG}}^2 \times D_{\text{crit}}$)

- **Inflation e-folds** (this paper)

The SAME integer $A_5 = 60$ governs biology, cosmology expansion rate, LENR, first stars, BH formation, AND the number of inflation e-folds that produced our observable universe.

NOT REPLACEMENT

Empirical inflation measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1462
- Related: PAPER_1666-1675 (tier-25); PAPER_1656-1665 (tier-24)
- Calculator dispatch: `calculate_paradox({"paradox": "horizon_60_efolds"})`

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